

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458490.8379 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.048
Transit depth (Rp/Rs)^2	2.54 +/- 1.51 [%]
Orbital Inclination (inc)	87.24 +/- 2.9
Airmass coefficient 1 (a1)	2.4 +/- 1.73
Airmass coefficient 2 (a2)	-0.53 +/- 0.83
Scatter in the residuals of the lightcurve fit is	89.22 %
Best Comparison Star	#1 - [234, 164]
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.091 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.159 $\pm$ 0.048 vs 0.1368 (deviation 0.3 σ)
Duration fit vs published	0.091 d vs 0.0758 d (deviation 0.98 σ)
Mid-time O-C	7.7 min
Depth SNR	2.1
Residual scatter	89.22 %

Light curve

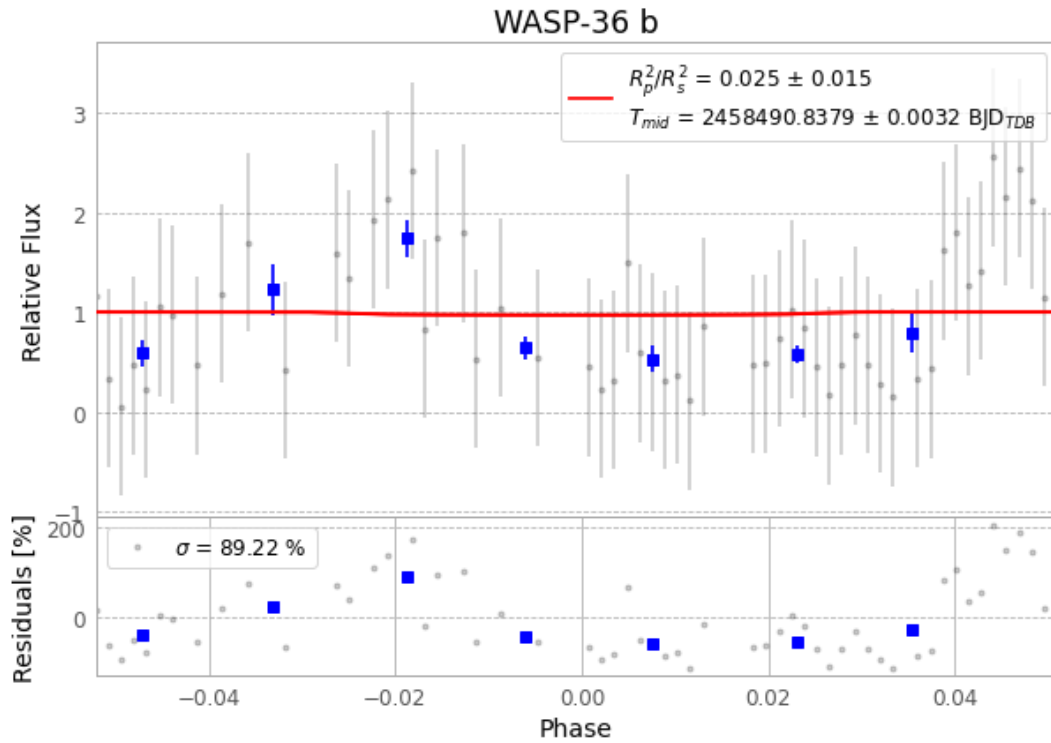


Figure 1 — FinalLightCurve_WASP-36 b_2019-01-06.png (SHA-256 2caa4112bbf97fc5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [503, 372], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fdf73583074bad8d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36190107060312.FITS → WASP-36190107095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190107.log (a7b3eb3469c6...), FinalLightCurve_WASP-36 b_2019-01-06.png (2caa4112bbf9...), FinalLightCurve_WASP-36 b_2019-01-06.csv (b670bb4e4cfa...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:18Z.